

Ali Awada

AI Engineer • Generative AI

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PROFESSIONAL SUMMARY

AI Engineer specialized in **generative AI solutions** and Computer Engineer (M.Eng., Concordia University) focused on Artificial Intelligence and Software Engineering. Experienced in building LLM-powered applications, AI agents and workflow automation with **Claude, MCP and n8n**, and intelligent backend systems in Python and Django. Sole developer of a **transformer model for human activity recognition** (88.5% weighted F1), compressed for edge deployment. Committed to reliable, secure and responsible AI with human-validated outputs.

TECHNICAL SKILLS

LLMs & Generative AI: LLMs • Generative AI • AI Agents • Agentic AI • Prompt Engineering • Conversational AI • NLP

AI Automation: Claude AI • MCP • n8n • API Integration (VirusTotal, AbuseIPDB)

Deep Learning & ML: Transformers • TensorFlow / Keras • TensorFlow Lite • Model Compression • Knowledge Distillation • PyTorch • scikit-learn • Logistic Regression • TF-IDF • Jupyter

ML Foundations: Statistics • Linear Algebra • Supervised Learning • Data Preprocessing • Feature Engineering • Model Evaluation & Metrics

Languages: Python • Java • C/C++ • JavaScript • SQL • PHP • Bash

Backend & Web: Django • REST APIs • PostgreSQL • React • Streamlit • OOP • Design Patterns

Tooling & Cloud: Git / GitHub • VS Code • Linux / Unix • Maven • JUnit • AWS, Azure, GCP (fundamentals) • CI/CD • MLOps concepts

Responsible AI: Human-in-the-Loop • AI Output Validation • Privacy & Security • AI Ethics

PROFESSIONAL EXPERIENCE

Research Associate – Human Activity Recognition

January 2026 – April 2026

[Concordia University, Montreal](#) • Sole developer • Supervised by Dr. Ahmed Bali & Dr. Paula Logo • [GitHub](#)

- Developed a transformer-based model (HART) that classifies 1.28 s of wrist IMU data into 9 activities on the PAMAP2 dataset and maps them to WHO intensity levels.
- Raised weighted F1 from 84.6% to **88.5%** by adding a multi-scale convolutional tokenizer and tuning learning rate and EMA decay.
- Built a knowledge-distillation pipeline that made the model **1.18x smaller** (9.14 → 7.73 MB) for an NVIDIA Jetson Orin Nano target, with TensorFlow Lite export.
- Fixed test-set leakage and 21 bugs across five files; made every run reproducible with JSON configs, fixed seeds and per-run metadata.

Backend Developer

January 2025 – April 2025

Concordia University, Montreal

- Developed backend services with Python and Django on a PostgreSQL relational database.
- Integrated AI chatbot capabilities into a web application.
- Applied Git, object-oriented programming and design patterns in a multidisciplinary team.

IT Support Technician

March 2023 – August 2023

Online Money Transfer (OMT), Lebanon

- Delivered Tier 1 support for hardware and software issues; installed, configured and maintained Windows workstations.
- Diagnosed hardware failures, supported local network infrastructure and kept the business running.

PROJECTS

Spam Message Detector

Oct 2026

Independent project • Sole developer • [GitHub](#)

- Built an SMS spam classifier (TF-IDF + Logistic Regression) reaching **97.58% accuracy and 89.3% spam recall** on ~5,200 messages, handling class imbalance with class weighting and threshold tuning.
- Deployed it as an interactive Streamlit app with real-time predictions, word-level explanations and batch CSV classification.

Secure Backend Web Application

2025

Concordia University • Django, PostgreSQL, Git

- Django backend with a PostgreSQL relational database, an integrated AI chatbot and version-controlled deployment, built in a multidisciplinary team.

Robot Motion Simulator

COEN 6761

Concordia University team project • Java 17, Maven, JUnit 5 • [GitHub](#)

- Command-driven robot on an $N \times N$ grid with movement, rotation, pen up/down drawing and history replay, covered by JUnit 5 tests.

EDUCATION

Master of Engineering – Electrical & Computer Engineering (Co-op)

2024 – 2026

Concordia University, Montréal, QC

- Specialization: Artificial Intelligence & Machine Learning, Software Programming, Networking & Cybersecurity.
- Key courses: ENGR 6971 Project and Report I (edge activity recognition), COEN 6761 Software Testing.

Bachelor of Science – Computer & Communication Engineering

2019 – 2022

Lebanese International University, Beirut, Lebanon

- GPA 3.24 / 4.00, **with honors**. Relevant courses: Java Programming I & II, Databases, Networking, Unix, Digital Logic.

LANGUAGES

Arabic (Native) • English (Fluent) • French (Intermediate)